and $\rho(\Lambda^{11}W^{11}) < 1$ by Theorem 3.2. Hence the eigenvalues on the unit circle, which decide convergence and asymptotic stability (Theorem 1.9), can only come from W^{22}, i.e. from the FDG model of the P-independent agents: the model is asymptotically stable when there are no P-independent agents ($r = n$), and it converges when the eigenvalue 1 of W^{22} is semi-simple and its other eigenvalues lie inside the unit disk, i.e. when W^{22} is regular.

3.1.3 How the opinions converge: the matrix V and the augmented graph

When all agents are P-dependent, the final opinions are given by (9), and the convex combination of Theorem 3.2 involves only the prejudices:

$$x_i(\infty) = \sum_{j=1}^{n} v_{ij}\, u_j.$$

Three facts follow.

- The initial opinions are forgotten: $x(0)$ does not appear in $x(\infty)$.
- Only the prejudices of prejudiced agents count: if $\lambda_j = 1$, the j-th column of $I - \Lambda$ is zero, hence so is the j-th column of V, and u_j has no weight in any final opinion.
- Every final opinion lies between the smallest and the largest prejudice of the prejudiced agents.

The comparison with the FDG model is instructive. There, the consensus value $w^\top x(0)$ is a weighted average of the initial opinions, with the same weights w for every agent (Theorem 2.1); here, the final opinions are weighted averages of the prejudices, and each agent has its own weights, given by its row of V.

Consensus and disagreement. In the FDG model, consensus is a property of the model: the model reaches a consensus if the final opinions coincide for any initial condition $x(0)$ [5]. In the FJ model with all agents P-dependent, the limit $x(\infty) = Vu$ does not depend on $x(0)$ (equation (9)): whether the final opinions coincide is decided by the prejudices u, which are an external input, through the matrix V, and no longer by the initial opinions. Accordingly, [5] studies the FJ model in terms of convergence and of the form of the limit, and not in terms of consensus.

The question whether the final opinions coincide can still be asked, but now it concerns the prejudices: for a given u, we say that the FJ model reaches a *consensus* if $x(\infty) = c\,\mathbf{1}_n$ for some $c \in \mathbb{R}$, and a *disagreement* otherwise. The answer comes from the equilibrium equation of Section 3.1.2. Since $W\mathbf{1}_n = \mathbf{1}_n$, a consensus vector $c\,\mathbf{1}_n$ is an equilibrium of (4) if and only if

$$c\,\mathbf{1}_n = \Lambda W c\,\mathbf{1}_n + (I - \Lambda)\,u \quad \iff \quad (I - \Lambda)\,(c\,\mathbf{1}_n - u) = 0,$$

i.e. $(1 - \lambda_i)(c - u_i) = 0$ for every agent i. For an agent that is not prejudiced ($\lambda_i = 1$) the condition is always satisfied; for a prejudiced agent it requires $u_i = c$. When all agents are P-dependent, the equilibrium is unique and it is the limit Vu (Corollary 3.3). Hence:

> *when all agents are P-dependent, the FJ model reaches a consensus if and only if all the prejudiced agents have the same prejudice, and the group agrees on it.*

In social terms, the network decides how much each prejudice weighs, through V, but the group can agree only if its anchored agents agree. Section 3.2.2 shows both cases on the same networks: a single prejudice, on which the group agrees, and different prejudices, which keep it divided.

The notion of consensus of the FDG model applies again only to the P-independent agents ($0 < r < n$), which evolve as an FDG model with matrix W^{22} and whose limit depends on their initial opinions $x^2(0)$ (Section 3.1.2). The P-dependent agents converge to convex combinations of the prejudices and of the final opinions of the P-independent agents (Theorem 3.2), so they do not necessarily end up at the consensus value of the P-independent agents, even when there is one.

The limit (9) has a direct interpretation in terms of the FDG model. Following [5], the FJ model (4) may be considered as an augmented FDG model with n virtual stubborn agents, anchored at the